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Hybrid Offline-First HMS: Project Scope

1B. Core Operational Framework & Architectural Proof-of-Concept

- 3-4 MONTH MVP
- PROJECT DURATION: 3 to 4 Months Max
- TEAM CAPACITY: 4 Student Engineers
- TARGET PLATFORM: Windows .exe + Embedded LAN
- CORE STRATEGY: Core Framework MVP

1. SCOPE RATIONALE: ENTERPRISE SCALE VS. STUDENT MVP

Commercial Hospital Management Systems (HMS) support complex ecosystems (intensive care units, multi-tier billing, operating theater scheduling, insurance claims, and hundreds of staff roles). Attempting a full-scale commercial clone in 3–4 months with 4 developers would lead to feature creep and unstable code. Our strategy is to focus on a high-reliability core operational framework that proves the technical feasibility of offline-first local execution, LAN display streaming, and background cloud sync. The proposed architecture is intentionally modular so that additional hospital modules such as laboratory management, appointment scheduling, billing, insurance processing, and electronic medical records can be integrated in future versions without redesigning the existing system. The MVP targets small clinics, diagnostic centers, and medium-sized hospitals where lightweight deployment, low hardware requirements, and uninterrupted local operation are more valuable than enterprise-scale functionality.

2. SCOPE BOUNDARY (IN-SCOPE VS. SCALED-DOWN COMPLEXITY)

DOMAIN / MODULE	FULL REAL-WORLD HMS SCALE	OUR PROJECT MVP SCOPE (IN-SCOPE)
User Roles & Access	Excluded: Surgeons, Nurses, Lab Techs, Insurance Admins, Accountants.	In-Scope: Dual-role Interface : Receptionist (Data entry/queue) & Doctor (Lookup/status).
Patient & Queueing	Excluded: Multi-specialty triage, ICU monitoring, surgical schedules.	In-Scope: Fast patient intake, token generation, and real-time OPD queue status.

Beds & Resources	Excluded: Ward-level wing mapping, equipment tracking, sanitation logs.	In-Scope: General bed matrix (Available / Occupied / Maintenance) with 1-click toggles.
Pharmacy & Inventory	Excluded: 1,000+ SKU inventory, batch expiry alerts, supplier purchase orders.	In-Scope: Core medicine stock ledger (~20 essential items), search query, low-stock warning.
Display Systems	Excluded: Complex digital signage software suites over cloud.	In-Scope: Embedded Client (Raspberry Pi/ ESP32) listening on local LAN for live OPD queues.
Data Architecture	Excluded: Enterprise server cluster, multi-master replication.	In-Scope: Single local database (SQLite) + Periodic background cloud sync API.

3. CORE TECHNICAL DELIVERABLES & ARCHITECTURE

A. Windows Desktop App (.exe) • High-speed UI (PyQt / C# WPF) for • Reception & Doctors. • Instant local database writes (<10ms latency). • Zero-internet fallback operating mode.	**B. Embedded Display Node** • Lightweight client on Pi / ESP32 + HDMI. • Broadcasts OPD queue & doctor status. • Direct local LAN discovery (No cloud dependency).	**C. Cloud Sync & Conflict Engine** • Background worker thread for auto-sync. • Rest API backend (FastAPI / Node) + PostgreSQL. • Deterministic conflict handler (Last-Write-Wins).

4. Functional Objectives

- **Patient Registration:** Enable quick and accurate registration of new and returning patients.
- **OPD Queue Management:** Generate and manage patient tokens with real-time queue updates.
- **Doctor Management:** Display doctor availability and allow patient lookup for consultations.
- **Bed Management:** Monitor and update the status of hospital beds (Available, Occupied, Maintenance).
- **Pharmacy Inventory:** Track medicine stock levels and provide low-stock notifications.
- **Role-Based Access:** Provide separate interfaces and permissions for receptionists and doctors.

- **Local Database Management:** Store and retrieve operational data efficiently using a local database.
- **Offline Operation:** Ensure uninterrupted hospital operations without requiring an internet connection.

5. 3.5-MONTH DEVELOPMENT ROADMAP (4 ENGINEERS)

Month 1: Core	Month 2: LAN & Embedded	Month 3: Cloud Sync Engine	Month 4 (0.5 Mo): Testing & Polish
• Desktop & DB • Setup SQLite local database schema. • Build Receptionist GUI (Patient entry & beds). • Build Doctor GUI (Schedule & lookups).	• Implement local LAN socket / MQTT broadcast. • Develop lightweight Embedded Display UI. • Test real-time queue streaming over local Wi-Fi.	• Develop Cloud API & remote database. • Build background sync worker with retry logic. • Implement conflict resolution for stock/beds.	• Simulate network drops & stress test offline mode. • Compile standalone `.exe` package. • Final documentation & demo setup.

6. Conclusion

The proposed Hybrid Offline-First Hospital Management System focuses on delivering a reliable and practical MVP within the project's time and resource constraints. By prioritizing core hospital operations, offline functionality, LAN-based communication, and cloud synchronization, the system demonstrates a scalable and efficient architecture. This approach provides a strong foundation for future expansion into a comprehensive Hospital Management System while ensuring successful implementation within the academic timeline.